

年度语料索引、方法与参考文献

可追溯的论文、博客、仓库清单, 纳入标准, 证据分级与研究局限

研究范围	2025-08-20 至 2026-08-19; 另列少量边界基线
语料规模	完整索引: 70篇论文, 45篇博客, 36个仓库
证据原则	优先论文原文、作者项目页、官方工程博客和GitHub仓库; 主张与推断分开
版本	v1.0 生成日期 2026-08-19

本报告不是排行榜。它是一套研究地图:
解释哪些结果可以相信、哪些只是方向性证据、哪些工程选择会改变结论。

本卷导航

01	研究问题与范围
02	检索、筛选与证据方法
03	语料统计与覆盖缺口
04	论文索引
05	官方/作者工程博客索引
06	GitHub仓库索引
07	局限、复核和更新建议

1. 研究问题与范围

严格时间窗为 2025-08-20 至 2026-08-19。主题范围是基于大语言模型或多模态模型的数字Agent，包括工具调用、规划、状态与记忆、多智能体、编码、Web/GUI、训练、评测、可靠性与安全。机器人和纯强化学习Agent 只在与LLM Agent架构直接相关时纳入。

核心研究问题

- 过去一年Agent研究的主要技术路线、实证转折和工程共识是什么？
- 论文、工程博客和开源仓库分别提供什么证据，彼此在哪里冲突？
- 哪些改进来自模型，哪些来自harness、环境、工具、记忆或grader？
- 长时、多工具、多Agent和高权限系统出现了哪些新失败模式？
- 怎样建立可复现、可诊断、成本归一化和安全相关的评测？
- 未来最值得投入的基础设施、研究问题和组织能力是什么？

边界基线

少量发表于窗口前数天或数月、但被窗口内工作大量继承的资料以“边界基线”保留，例如Agent Lightning、tau2-bench、BrowseComp-Plus、A2A发布和Anthropic多Agent研究系统。它们不计入“窗口内新成果”判断，只用于解释技术路径。

2. 检索、筛选与证据方法

阶段	操作	目的
发现	按月份和主题检索arXiv、官方研究/工程博客、GitHub	广覆盖候选
核验	回到论文摘要/原文、作者项目页、官方博客和仓库API	避免二次报道误差
纳入	明确Agent对象、在窗口内或显式边界、提供方法/系统/实证价值	排除纯营销和重复转述
编码	日期、类型、主题、主要贡献、限制、原始链接	统一比较
综合	按系统层、任务、证据强度和失败因果链归纳	不按热度简单排序
复核	交叉检查关键数字、时间窗、动态仓库元数据和边界标签	降低事实错误

纳入标准

- 论文: 提出新架构、训练方法、记忆/技能机制、基准、安全/可靠性分析或系统综述。
- 博客: 由模型/平台开发者、研究机构或仓库维护者发布，包含可复用工程细节、测量或事故经验。
- 仓库: 具有明确Agent相关定位、公开代码/规范，并能代表一种生态层或重要应用。
- 排除: 只有概念宣传、没有可核验原始来源、与LLM Agent关系弱、重复内容或明显低质量聚合。

证据分级

级别	含义	例子
A	可执行确定性验证	测试、状态检查、容器化重放
B	人工审计/专家复核	盲审、故障标注、数据质量审计
C	受控实验与消融	固定模型、工具和预算的组件比较
D	真实使用/运营观察	匿名化使用数据、事故报告、生产记录
E	作者或厂商主张	产品发布指标、演示、客户案例
F	本系列推断	跨来源综合与趋势判断

3. 语料统计与覆盖缺口

语料	总数	严格窗口	说明
论文/预印本	70	60	arXiv为主; 含基准、方法、综述
官方/作者博客	45	43	Anthropic/OpenAI/Microsoft/Google/HF为主
GitHub仓库	36	36	元数据为动态快照; 项目创建日可早于窗口

覆盖较强

- 长时任务、工具/MCP、记忆、自演化、编码、GUI、基准与安全。
- 北美与中国主要开源Agent项目及大型研究机构工程博客。
- 从模型调用到harness、执行、验证和治理的系统层。

覆盖较弱

- 闭源企业Agent的真实故障率、成本和内部架构，公开信息有限。
- 非英语社区、较小公司和行业私有仓库可能被低估。
- 论文引用数和GitHub stars在2026-08仍快速变化，不适合稳定排序。
- 部分2026新论文尚未经过正式同行评审或独立复现。
- 动态Web和GUI基准难以完全复现同一环境。

4. 论文索引

P01 | 2025-08 | long-horizon/eval | 窗口内
OdysseyBench: Evaluating LLM Agents on Long-Horizon Cross-Application Workflows
主要贡献: 300 real-use tasks plus 302 synthesized office tasks expose cross-application state and dependency failures.
阅读限制: Synthetic tasks and application versions can bias difficulty; report partial progress, not only completion.
[原文](#)

P02 | 2025-08 | tools/MCP | 窗口内

MCPVerse: An Expansive, Real-World Benchmark for Agentic Tool Use

主要贡献: More than 550 executable tools create a tool-description space above 140k tokens and stress routing at scale.

阅读限制: Live tools improve realism but reduce longitudinal reproducibility.

[原文](#)

P03 | 2025-08 | tools/MCP | 窗口内

MCP-Bench: Benchmarking Tool-Using LLM Agents with Real MCP Servers

主要贡献: 28 live servers and 250 tools test cross-tool planning and parameter control rather than isolated function calling.

阅读限制: Environment drift and secret/API availability complicate replication.

[原文](#)

P04 | 2025-08 | tools/MCP | 窗口内

MCP-Universe: Benchmarking Models with Real-World MCP Servers

主要贡献: Six domains and 11 servers turn protocol compliance into an end-to-end operations test.

阅读限制: Coverage is broad but each domain remains shallow compared with production estates.

[原文](#)

P05 | 2025-08 | tools/MCP | 边界基线

LiveMCPBench: Can Agents Navigate an Ocean of MCP Tools?

主要贡献: 70 servers and 527 tools show that tool routing and outcome verification become separate bottlenecks.

阅读限制: LLM-as-judge outcome checks need calibration against deterministic state when possible.

[原文](#)

P06 | 2025-08 | security/MCP | 窗口内

MCPTox: Tool Poisoning Attacks on Real MCP Servers

主要贡献: 45 servers, 353 tools and 1,312 malicious cases demonstrate that tool metadata itself is an attack surface.

阅读限制: Attack success depends on model, tool descriptions and permission policy; do not generalize one aggregate score.

[原文](#)

P07 | 2025-08 | domain/eval | 边界基线

Finance Agent Benchmark

主要贡献: Recent SEC-filing tasks test research, calculation, provenance and synthesis under time-sensitive evidence.

阅读限制: Finance answers can be verifiable while investment conclusions remain judgment-dependent.

[原文](#)

P08 | 2025-08 | computer-use/eval | 边界基线

NaturalGAIA: Verifiable Long-Horizon GUI Tasks

主要贡献: A hierarchical framework and verifiable tasks connect GUI action sequences to measurable end states.

阅读限制: The benchmark may reward its own hierarchy; cross-harness replication is needed.

[原文](#)

P09 | 2025-08 | planning/RL | 边界基线

PilotRL: Global Planning-Guided Training for Language Agents

主要贡献: Progressive RL teaches adherence to explicit global plans before autonomous execution.

阅读限制: Benefits can depend on plan quality; rigid plans may hurt dynamic recovery.

[原文](#)

P10 | 2025-08 | planning | 窗口内

HiPlan: Hierarchical Planning with Adaptive Global-Local Guidance

主要贡献: Milestone libraries and local hints reuse demonstrations without placing complete trajectories in context.

阅读限制: Retrieval quality and domain similarity bound transfer.

[原文](#)

P11 | 2025-08 | tools/MCP | 边界基线

MCPToolBench++: Large-Scale MCP Tool Calling

主要贡献: Large multi-domain tool pools quantify the context cost of exposing schemas to the model.

阅读限制: Tool-call correctness is necessary but not sufficient for task success.

[原文](#)

P12 | 2025-08 | web/domain | 边界基线

ShoppingBench: Intent-Grounded End-to-End Shopping Agents

主要贡献: Budget, vouchers and multi-product constraints move shopping evaluation beyond search-only tasks.

阅读限制: Dynamic catalogs and commercial interfaces can quickly invalidate results.

[原文](#)

P13 | 2025-08 | coding/eval | 窗口内

GitTaskBench: Code Agents on Real Repository Workflows

主要贡献: 54 tasks across seven modalities test repository use beyond patch generation.

阅读限制: Small task count limits fine-grained model ranking.

[原文](#)

P14 | 2025-08 | planning/eval | 边界基线

HeroBench: Long-Horizon Hierarchical Planning in Virtual Worlds

主要贡献: Hundreds to thousands of actions expose resource-dependency and global consistency failures.

阅读限制: Virtual-world structure is diagnostic but not a direct proxy for office or coding work.

[原文](#)

P15 | 2025-09 | reliability | 窗口内

Where LLM Agents Fail and How Errors Cascade

主要贡献: Root-cause errors propagate through planning, memory, reflection and tool modules; final failure labels hide the first decisive mistake.

阅读限制: Taxonomies are sensitive to annotator access to hidden state and intended plans.

[原文](#)

P16 | 2025-09 | long-horizon/eval | 窗口内

Benchmarking Agent Capabilities in Ultra Long-Horizon Tasks

主要贡献: Heavy settings exceed 200k tokens and 400 tool calls; simple scale fails under in-context locking and capability gaps.

阅读限制: One environment cannot settle how general the identified failure causes are.

[原文](#)

P17 | 2025-09 | interaction/eval | 窗口内

VitaBench: Versatile Interactive Tasks in Real-World Applications

主要贡献: Tests extensive information, diverse resources and dynamic user interaction in one suite.

阅读限制: Breadth can dilute causal diagnosis unless component-level metrics are retained.

[原文](#)

P18 | 2025-09 | training | 窗口内

AgentScaler: General Agentic Intelligence via Environment Scaling

主要贡献: Automatically generated heterogeneous environments support staged function-calling training and domain specialization.

阅读限制: Simulated environment diversity is not equivalent to deployment diversity.

[原文](#)

P19 | 2025-09 | coding/security | 窗口内

SecureAgentBench: Secure Code Generation in Real Repositories

主要贡献: Best evaluated setup achieves only 15.2% correct-and-secure solutions, showing that functional tests miss security regressions.

阅读限制: 105 tasks are valuable but insufficient to cover language and vulnerability diversity.

[原文](#)

P20 | 2025-09 | coding/eval | 窗口内

PerfBench: Can Agents Resolve Real-World Performance Bugs?

主要贡献: Performance repairs require agents to create measurements and beat developer baselines, not merely pass tests.

阅读限制: Hardware variance and noisy benchmarks can dominate small improvements.

[原文](#)

P21 | 2025-09 | planning/RL | 窗口内

Planner-R1: Reward Shaping for Agentic RL with Smaller Models

主要贡献: Structured reward shaping improves planning and tool-use learning with limited training samples.

阅读限制: Reward design may encode the benchmark solution and reduce open-world transfer.

[原文](#)

P22 | 2025-09 | training/tools | 窗口内

CodeGym: Generalizable End-to-End Tool-Use RL

主要贡献: Executable code environments generate diverse, verifiable workflows for agent RL.

阅读限制: Coding is a useful substrate but underrepresents ambiguous human and organizational constraints.

[原文](#)

P23 | 2025-10 | memory/multi-agent | 窗口内

LEGOMem: Modular Procedural Memory for Multi-Agent Workflows

主要贡献: OfficeBench results suggest orchestrator memory helps decomposition while task-agent memory improves execution accuracy.

阅读限制: Placement effects may shift with topology, model size and retrieval policy.

[原文](#)

P24 | 2025-10 | planning/context | 窗口内

ReCAP: Recursive Context-Aware Reasoning and Planning

主要贡献: Recursive subgoals and context updates improve alignment in long-horizon environments.

阅读限制: Recursion adds model calls and may amplify evaluator error.

[原文](#)

P25 | 2025-10 | memory/self-evolution | 窗口内

MUSE: Experience-Driven Self-Evolving Agents

主要贡献: Structured reflections are integrated into hierarchical memory after each subtask for later planning.

阅读限制: Self-written memories can preserve mistaken causal stories unless externally verified.

[原文](#)

P26 | 2025-10 | self-evolution | 窗口内

ECHO: Sample-Efficient Online Learning in Language Agents

主要贡献: Experience reuse improves adaptation in navigation and collaborative information gathering.

阅读限制: Reported gains depend on stateful benchmark construction and comparable memory budgets.

[原文](#)

P27 | 2025-10 | security/survey | 窗口内

Agentic AI Security: Threats, Defenses and Evaluation

主要贡献: Organizes prompt injection, tool abuse, memory attacks and governance into a systems threat model.

阅读限制: Survey evidence mixes proofs, synthetic attacks and deployment reports of unequal strength.

[原文](#)

P28 | 2025-10 | tools/eval | 窗口内

TRAJECT-Bench: Trajectory-Aware Tool-Use Evaluation

主要贡献: Separates tool selection, arguments, dependency order and final answer to expose parameter-blind choices.

阅读限制: Reference trajectories are not always unique; outcome-equivalent alternatives require careful grading.

[原文](#)

P29 | 2025-11 | memory/eval | 窗口内

Evo-Memory: Test-Time Learning with Self-Evolving Memory

主要贡献: A streaming benchmark compares more than ten memory modules across task sequences and makes update policy measurable.

阅读限制: Cross-dataset aggregation can hide incompatible memory objectives.

[原文](#)

P30 | 2025-11 | tools/routing | 窗口内

Tool-to-Agent Retrieval for Large Tool Ecosystems

主要贡献: Routes at tool granularity rather than coarse agent descriptions, reducing capability loss behind delegation layers.

阅读限制: Retrieval errors still compound with downstream agent errors.

[原文](#)

P31 | 2025-11 | tools/planning | 窗口内

TPS-Bench: Tool Planning and Scheduling

主要贡献: 200 compounding tasks distinguish sequential completion from parallel scheduling efficiency; small-sample RL improves both time and completion.

阅读限制: Timing depends on infrastructure and service latency, not only policy.

[原文](#)

P32 | 2025-11 | multi-agent/security | 窗口内

TAMAS: Adversarial Risks in Multi-Agent LLM Systems

主要贡献: Demonstrates that adversarial influence can propagate across role and message boundaries.

阅读限制: Threat success rates are topology- and defense-specific.

[原文](#)

P33 | 2025-11 | multi-agent/reliability | 窗口内

Detecting Silent Failures in Multi-Agent Trajectories

主要贡献: Treats silent failure as trajectory anomaly detection rather than final-answer error.

阅读限制: Detection does not automatically yield correct attribution or recovery.

[原文](#)

P34 | 2025-11 | memory/training | 窗口内

Trainable Graph Memory for Agent Strategy

主要贡献: Structured decision paths are distilled into meta-cognition whose weights are updated with reward feedback.

阅读限制: Graph abstractions can discard rare but decisive context.

[原文](#)

P35 | 2025-12 | memory/survey | 窗口内

Memory in the Age of AI Agents

主要贡献: Distinguishes token, parametric and latent forms; factual, experiential and working functions; and formation, evolution and retrieval dynamics.

阅读限制: A taxonomy organizes the field but does not establish which design wins under equal budgets.

[原文](#)

P36 | 2025-12 | memory/architecture | 窗口内

Hindsight: Memory that Retains, Recalls and Reflects

主要贡献: Separates facts, experiences, summaries and beliefs; reported LongMemEval accuracy rises from 39% to 83.6% with the same 20B backbone.

阅读限制: Conversational memory gains may not transfer to action-heavy coding and GUI tasks.

[原文](#)

P37 | 2025-12 | eval/reliability | 窗口内

Beyond Task Completion: Agent Assessment Framework

主要贡献: Four pillars cover model, memory, tools and environment, revealing runtime deviations missed by binary success.

阅读限制: One CloudOps validation case limits generalization.

[原文](#)

P38 | 2025-12 | data-science/tools | 窗口内

ML-Tool-Bench: Tool-Augmented Planning for ML Tasks

主要贡献: Deterministic shaped feedback and decomposition outperform tree search when state scoring is inconsistent.

阅读限制: Only 15 Kaggle challenges and in-memory state reduce production comparability.

[原文](#)

P39 | 2025-12 | coding/architecture | 窗口内

Confucius Code Agent: Industrial-Scale Open Coding Agent

主要贡献: Combines hierarchical working memory, persistent notes and a build-test-improve meta-agent; reports 54.3% on SWE-Bench Pro.

阅读限制: Leaderboard claims require reruns under controlled infrastructure and clean tasks.

[原文](#)

P40 | 2026-01 | memory/architecture | 窗口内

E-mem: Episodic Context Reconstruction

主要贡献: Assistant agents keep uncompressed local contexts while a master orchestrates; reported cost falls over 70% on LoCoMo.

阅读限制: LoCoMo emphasizes dialogue; master-assistant costs and error propagation need action-domain tests.

[原文](#)

P41 | 2026-01 | research/eval | 窗口内

DeepSearchQA: Comprehensiveness for Deep Research Agents

主要贡献: 900 prompts in 17 fields test exhaustive collection, de-duplication, entity resolution and stopping.

阅读限制: Open-web ground truth and answer sets require continual maintenance.

[原文](#)

P42 | 2026-01 | tools/reliability | 窗口内

When Agents Fail to Act: Tool Invocation Reliability

主要贡献: 1,980 deterministic tests across models and hardware classify 12 procedural error types.

阅读限制: Tool initialization results may not transfer to complex reasoning-heavy workflows.

[原文](#)

P43 | 2026-02 | tools/eval | 窗口内

MCP-Atlas: Real MCP Server Tool Competency

主要贡献: 36 servers, 220 tools and 1,000 tasks use claims-based partial credit and internal diagnostics.

阅读限制: Claims scoring still needs validation against actual external state changes.

[原文](#)

P44 | 2026-02 | security/eval | 窗口内

AgentLAB: Long-Horizon Attacks

主要贡献: 644 cases across 28 environments cover intent hijacking, tool chaining, task injection, objective drift and memory poisoning.

阅读限制: Attack realism depends on whether adversaries can actually reach those channels in deployment.

[原文](#)

P45 | 2026-02 | memory/skills | 窗口内

MemSkill: Learning and Evolving Memory Skills

主要贡献: A controller selects memory routines while a designer revises the skill set from hard cases.

阅读限制: Designer-generated skills risk self-reinforcing benchmark heuristics.

[原文](#)

P46 | 2026-02 | skills/survey | 窗口内

Agent Skills for LLMs: Architecture, Acquisition and Security

主要贡献: Treats skills as progressively disclosed packages and proposes lifecycle governance for provenance and permissions.

阅读限制: Security prevalence estimates depend heavily on repository sampling and vulnerability definitions.

[原文](#)

P47 | 2026-02 | industry/transparency | 窗口内

The 2025 AI Agent Index

主要贡献: Documents 30 agents across 45 fields and reports major gaps in public safety and design disclosure.

阅读限制: Public documentation and developer correspondence cannot reveal hidden production controls.

[原文](#)

P48 | 2026-02 | coding/eval | 窗口内

OmniCode: Broad Software Engineering Agent Evaluation

主要贡献: 1,794 tasks span bug fixes, test generation, review fixes and style work in Python, Java and C++.

阅读限制: Synthetic and recent tasks improve contamination resistance but change ecological realism.

[原文](#)

P49 | 2026-03 | coding/long-horizon | 窗口内

SWE-CI: Maintaining Codebases through Continuous Integration

主要贡献: 100 tasks average 233 days and 71 commits, shifting evaluation from one patch to maintainability across evolution.

阅读限制: Reconstructing historical CI environments is costly and may introduce infrastructure noise.

[原文](#)

P50 | 2026-03 | security/survey | 窗口内

Clawed and Dangerous: Trusting Open Agentic Systems

主要贡献: A six-dimensional taxonomy argues that behavior governance under uncertainty is the central security problem.

阅读限制: A systematization of knowledge should not be read as empirical proof of every control.

[原文](#)

P51 | 2026-04 | security/eval | 窗口内

ATBench: Realistic Trajectory Benchmark for Long-Horizon Safety

主要贡献: 1,000 audited trajectories organize risks by source, failure mode and harm with delayed triggers.

阅读限制: Average trajectories remain shorter than many production tasks.

[原文](#)

P52 | 2026-04 | computer-use/eval | 窗口内

WindowsWorld: Cross-Application Professional GUI Agents

主要贡献: 181 tasks across 17 apps show under 21% success on multi-application workflows and early stalling beyond three apps.

阅读限制: Windows configuration and automation interfaces materially affect results.

[原文](#)

P53 | 2026-05 | memory/eval | 窗口内

GroupMemBench: Memory in Multi-Party Conversations

主要贡献: Best system reaches only 46%; BM25 matches or beats many agent memories when speaker structure and terminology matter.

阅读限制: Synthetic group conversations may not capture organizational power and privacy constraints.

[原文](#)

P54 | 2026-05 | memory/eval | 窗口内

EvoMemBench: Memory from a Self-Evolving Perspective

主要贡献: Across 15 methods, long context remains competitive and no memory form wins all settings.

阅读限制: Benchmark parity depends on equalizing token, latency and storage budgets.

[原文](#)

P55 | 2026-05 | memory/eval | 窗口内

MemGym: Long-Horizon Memory Environment

主要贡献: Unifies tool dialogue, deep research, coding and computer use with memory-isolated scores.

阅读限制: Reward-model proxies can diverge from expensive end-to-end rollouts.

[原文](#)

P56 | 2026-05 | self-evolution | 窗口内

APEX: Autonomous Policy Exploration for Self-Evolving Agents

主要贡献: A strategy DAG balances exploration and exploitation to prevent memory-driven policy collapse.

阅读限制: Jericho and WebArena evidence is promising but narrow for cross-domain policy evolution.

[原文](#)

P57 | 2026-05 | memory/self-evolution | 窗口内

EvolveMem: Self-Evolving Retrieval Architecture

主要贡献: Failure logs drive guarded configuration changes with revert-on-regression and explore-on-stagnation.

阅读限制: Meta-optimization can overfit benchmark feedback unless held-out deployments are used.

[原文](#)

P58 | 2026-05 | multi-agent/architecture | 窗口内

Coordination as an Architectural Layer

主要贡献: Five controlled coordination configurations produce different calibration and cost signatures even with one fixed model.

阅读限制: n=100 results are directional and do not establish a universal best topology.

[原文](#)

P59 | 2026-05 | multi-agent/survey | 窗口内

Beyond Individual Intelligence: Collaboration, Attribution and Evolution

主要贡献: The LIFE progression links foundation, integration, failure attribution and self-evolution as one causal chain.

阅读限制: The framework is a research map, not a validated production lifecycle.

[原文](#)

P60 | 2026-06 | security/survey | 窗口内

Toward Secure LLM Agents: Threat Surfaces and Defenses

主要贡献: Synthesizes 247 papers around information flow, delegated authority and persistent state.

阅读限制: Fast-moving evidence means control effectiveness should be re-audited per release.

[原文](#)

P61 | 2026-06 | computer-use/eval | 窗口内

OSWorld 2.0: Long-Horizon Real-World Computer Use

主要贡献: 108 workflows take humans a median 1.6 hours; the best reported system completes only 20.6% at 500 steps.

阅读限制: Benchmark environments are expensive and sensitive to application state.

[原文](#)

P62 | 2026-06 | coding/interaction | 窗口内

Dialogue SWE-Bench: Dialogue-Driven Coding Agents

主要贡献: User simulation shows that better autonomous coding does not guarantee better clarification dialogue.

阅读限制: Simulator fidelity and persona assumptions can alter measured dialogue quality.

[原文](#)

P63 | 2026-07 | memory/long-horizon | 窗口内

AgenticSTS: Bounded-Memory Long-Horizon Testbed

主要贡献: Typed retrieval keeps prompts bounded across hundreds of decisions; skill-layer gains are reported as directional due to sample size.

阅读限制: Small controlled samples cannot support strong superiority claims.

[原文](#)

P64 | 2026-07 | reliability/survey | 窗口内

Beyond the Leaderboard: Tool, Planning and Reasoning Failures

主要贡献: Across 19 benchmarks, failures compound nonlinearly and more scaffolding does not consistently improve reliability.

阅读限制: Cross-paper taxonomies inherit inconsistent setups and metrics.

[原文](#)

P65 | 2026-07 | multi-agent/reliability | 窗口内

AgentLocate: Failure Localization in Multi-Agent Systems

主要贡献: Attributes both the responsible agent and earliest decisive step using multi-perspective verification.

阅读限制: LLM judges may share biases with the target system and need human calibration.

[原文](#)

P66 | 2026-08 | coding/reliability | 窗口内

Engineering Reliable Coding Agents

主要贡献: A multivocal system-level review treats evaluation, harness, state, permissions and infrastructure as a dependency chain.

阅读限制: Evidence quality varies across scholarly and practitioner records; practices are hypotheses until workload-tested.

[原文](#)

P67 | 2026-08 | skills/self-evolution | 窗口内

HyperSkill: Hypergraph-Structured Skill Memory

主要贡献: Links subtasks and reusable skills across trajectories with dual-path retrieval and structure-aware maintenance.

阅读限制: Very recent preprint; independent replication and cost-normalized ablations are not yet mature.

[原文](#)

P68 | 2025-08 | training/infrastructure | 边界基线

Agent Lightning: Train Any Agent with Reinforcement Learning

主要贡献: Decouples agent execution from RL training through a transition interface and credit assignment layer.

阅读限制: Published 15 days before the strict window; included as an explicit boundary baseline because it shaped the year's training stack.

[原文](#)

P69 | 2025-06 | interaction/eval | 边界基线

tau2-bench: Dual-Control Conversational Agents

主要贡献: Shared user-agent control causes large performance drops and exposes communication as a distinct capability.

阅读限制: Published before the strict window; retained only as a baseline used by later work.

[原文](#)

P70 | 2025-08 | research/eval | 边界基线

BrowseComp-Plus: Reproducible Deep-Research Evaluation

主要贡献: A fixed corpus and verified supports disentangle retrieval from reasoning and improve comparison fairness.

阅读限制: Published 12 days before the strict window; included as a boundary baseline.

[原文](#)

5. 官方/作者工程博客索引

B01 | 2025-09-11 | Anthropic | 窗口内

Writing effective tools for agents - with agents

Tool descriptions, outputs and error messages are part of the policy surface and should be optimized with task evals.

类型: tools/eval | [原文](#)

B02 | 2025-09-25 | OpenAI | 窗口内

GDPval: Measuring real-world economically valuable tasks

Moves beyond school-style questions toward occupation-grounded deliverables, while still relying on rubric and expert judging.

类型: eval/knowledge-work | [原文](#)

B03 | 2025-09-29 | Anthropic | 窗口内

Effective context engineering for AI agents

Treats context as a finite working set that must be curated, compressed, isolated and retrieved deliberately.

类型: context | [原文](#)

B04 | 2025-09-29 | Anthropic | 窗口内

Building agents with the Claude Agent SDK

Presents computer access, subagents, compaction and verification as general agent primitives beyond coding.

类型: harness | [原文](#)

B05 | 2025-10-06 | OpenAI | 窗口内

Introducing AgentKit

Bundles workflow design, UI, evaluation and optimization, while a 2026 update illustrates how fast agent platforms can be deprecated.

类型: platform | [原文](#)

B06 | 2025-10-16 | Anthropic | 窗口内

Equipping agents for the real world with Agent Skills

Skills use progressive disclosure to move reusable procedures out of monolithic prompts.

类型: skills | [原文](#)

B07 | 2025-10-20 | Anthropic | 窗口内

Beyond permission prompts: safer Claude Code sandboxing

Filesystem and network isolation reduce prompt fatigue while bounding impact; Anthropic reports 84% fewer permission prompts internally.

类型: security/sandbox | [原文](#)

B08 | 2025-10-21 | Microsoft Research | 窗口内

Tell me when: agents that wait, monitor and act

SentinelStep wraps dynamic polling with context management for hours- or days-long monitoring.

类型: long-running | [原文](#)

B09 | 2025-10-27 | Hugging Face | 窗口内

huggingface_hub v1.0 and tiny agents

Shows how an MCP client plus an agent loop can be small, shifting complexity to tool contracts and operations.

类型: tools/MCP | [原文](#)

B10 | 2025-11-04 | Anthropic | 窗口内

Code execution with MCP

Recommends code as a meta-tool so large tool estates are discovered and composed without loading every schema into context.

类型: tools/context | [原文](#)

B11 | 2025-11-04 | Microsoft Research | 窗口内

RedCodeAgent: automatic red-teaming of code agents

Combines memory, attack tools and iterative evaluation to discover unsafe code-agent behavior.

类型: security/coding | [原文](#)

B12 | 2025-11-24 | Anthropic | 窗口内

Prompt injection defenses for browser use

Treats every webpage as untrusted input and motivates layered defenses rather than prompt-only protection.

类型: security/prompt-injection | [原文](#)

B13 | 2025-11-24 | Microsoft Research | 窗口内

Fara-7B: efficient computer-use model

Argues that smaller on-device agents can improve latency and privacy but still require sandboxing and oversight.

类型: computer-use | [原文](#)

B14 | 2025-11-26 | Anthropic | 窗口内

Effective harnesses for long-running agents

Initializer, incremental feature work, durable artifacts and explicit end-to-end tests help cross-context progress.

类型: harness/long-running | [原文](#)

B15 | 2025-12-01 | Anthropic | 窗口内

AI agents find simulated smart-contract exploits

SCONE-bench links agent capability to economic impact and emphasizes sandboxed, reproducible execution.

类型: security/cyber | [原文](#)

B16 | 2025-12-04 | Hugging Face | 窗口内

Coding agents can run an open-model fine-tuning lifecycle

Demonstrates data checks, training submission, monitoring and conversion as one agent workflow.

类型: coding/science | [原文](#)

B17 | 2025-12-11 | Microsoft Research | 窗口内

Agent Lightning: adding RL without agent rewrites

Makes the agent runtime observable to a separate training system and reframes trajectories as credit-assignment data.

类型: training | [原文](#)

B18 | 2025-12-11 | Google | 窗口内

ADK with the Interactions API

Separates inner model loop from outer application loop, with server-side state and background execution.

类型: state/API | [原文](#)

B19 | 2025-12-15 | Google | 窗口内

A2UI: an open project for agent-driven interfaces

A declarative UI protocol constrains agent output to host-approved components instead of arbitrary executable UI.

类型: protocol/UI | [原文](#)

B20 | 2026-01-09 | Anthropic | 窗口内

Demystifying evals for AI agents

Defines the evaluated object as model plus harness and recommends transcript review, deterministic graders and representative tasks.

类型: eval | [原文](#)

B21 | 2026-01-23 | OpenAI | 窗口内

Unrolling the Codex agent loop

Makes prompt construction, inference and loop performance visible as engineering components.

类型: harness/coding | [原文](#)

B22 | 2026-02-05 | Anthropic | 窗口内

Quantifying infrastructure noise in coding evals

Shows infrastructure configuration can move scores by more than leaderboard gaps.

类型: eval/infrastructure | [原文](#)

B23 | 2026-02-05 | Anthropic | 窗口内

Building a C compiler with parallel Claude agents

A large autonomous experiment highlights parallelism gains, merge pressure and the need for testable interfaces.

类型: multi-agent/coding | [原文](#)

B24 | 2026-02-11 | OpenAI | 窗口内

Harness engineering in an agent-first repository

Repository rules, mechanical invariants and cleanup routines turn the codebase into durable context for future agents.

类型: coding/organization | [原文](#)

B25 | 2026-02-18 | Anthropic | 窗口内

Measuring AI agent autonomy in practice

Tail turn duration grows, but autonomy duration is not equivalent to correctness or value.

类型: usage/autonomy | [原文](#)

B26 | 2026-02-18 | OpenAI | 窗口内

Introducing EVMbench

Evaluates detection, patching and exploitation in a reproducible blockchain environment.

类型: security/eval | [原文](#)

B27 | 2026-02-26 | Microsoft Research | 窗口内

CORPGEN and multi-horizon workplace tasks

Under multi-task load, completion drops sharply; hierarchical planning, memory isolation and experience improve results.

类型: multi-task/office | [原文](#)

B28 | 2026-03-06 | Anthropic | 窗口内

Eval awareness in BrowseComp

A web-enabled model recognized and recovered benchmark answers, demonstrating that environment access changes contamination risk.

类型: eval/integrity | [原文](#)

B29 | 2026-03-19 | OpenAI | 窗口内

Monitoring internal coding agents for misalignment

Post-run monitoring classifies actions and severity; no true sabotage positives were reported as of 2026-03-17.

类型: security/monitoring | [原文](#)

B30 | 2026-03-23 | Anthropic | 窗口内

Long-running Claude for scientific computing

Applies persistent memory, test oracles and orchestration to multi-day scientific work.

类型: science/long-running | [原文](#)

B31 | 2026-03-24 | Anthropic | 窗口内

Harness design for long-running application development

Feature lists, artifacts and specialized passes improve sustained frontend and application work.

类型: harness/coding | [原文](#)

B32 | 2026-04-01 | Google | 窗口内

Developer guide to ADK agents with skills

Reports up to 90% token reduction from progressive disclosure and describes skill factories.

类型: skills/context | [原文](#)

B33 | 2026-04-08 | Anthropic | 窗口内

Scaling Managed Agents: decoupling brain from hands

Stable interfaces between harness and compute allow sandboxes and orchestration to evolve independently.

类型: runtime | [原文](#)

B34 | 2026-04-15 | OpenAI | 窗口内

The next evolution of the Agents SDK

Adds native sandbox execution and separates harness from compute for security, durability and scale.

类型: runtime | [原文](#)

B35 | 2026-04-23 | Anthropic | 窗口内

Claude Code quality postmortem

Three independent product changes created perceived quality regressions, illustrating system-layer confounding.

类型: operations | [原文](#)

B36 | 2026-04-27 | OpenAI | 窗口内

Symphony: open Codex orchestration spec

Publishes an orchestration specification so teams can implement tailored coding-agent supervisors.

类型: orchestration | [原文](#)

B37 | 2026-05-25 | Anthropic | 窗口内

How we contain Claude across products

Frames safety as blast-radius control across data, network, execution and account boundaries.

类型: security/containment | [原文](#)

B38 | 2026-06-08 | Anthropic | 窗口内

Paving the way for AI agents in biology

Even strong research agents fail to consistently construct accurate sequence datasets, emphasizing domain-specific verification.

类型: science/reliability | [原文](#)

B39 | 2026-06-16 | Anthropic | 窗口内

How Claude Code is used in practice

Usage data shows expanding adoption and higher-level delegation, but self-reported productivity is not causal proof.

类型: usage/coding | [原文](#)

B40 | 2026-06-30 | Microsoft Research | 窗口内

SkillOpt: agent skills as trainable parameters

Bounded edits, validation gates and rollback make skill text an auditable optimization layer.

类型: skills/optimization | [原文](#)

B41 | 2026-07-08 | OpenAI | 窗口内

Separating signal from noise in coding evaluations

A human-reviewed audit estimates about 30% of SWE-Bench Pro public tasks are broken.

类型: eval/data-quality | [原文](#)

B42 | 2026-07-30 | Microsoft Research | 窗口内

Echoverse: evolving computer-use environments

Argues that richer, changing environments can teach robust behavior better than simply adding static tasks.

类型: training/environment | [原文](#)

B43 | 2026-08-03 | Microsoft Research | 窗口内

Orchard: scalable agentic AI framework

A Kubernetes environment service supports data collection, RL rollouts and evaluation across harnesses.

类型: training/runtime | [原文](#)

B44 | 2025-06-13 | Anthropic | 边界基线

How we built a multi-agent research system

Parallel subagents and externalized phase summaries provide a pre-window baseline for later context-isolation designs.

类型: multi-agent/research | [原文](#)

B45 | 2025-04-09 | Google | 边界基线

Agent2Agent protocol announcement

Defines an interoperability layer between opaque agent applications; included as protocol baseline.

类型: protocol/A2A | [原文](#)

6. GitHub仓库索引

R01 | openai/openai-agents-python | SDK/runtime

快照: Python; MIT; 28,773 stars; created 2025-03; active

设计: Small agent loop with agents, tools, handoffs, sessions, guardrails and tracing.

适合: OpenAI-centric service agents and multi-agent workflows.

注意: Provider coupling and rapidly moving platform APIs require version pinning.

[GitHub](#)

R02 | openai/openai-agents-js | SDK/runtime

快照: TypeScript; MIT; 3,659 stars; created 2025-05; active

设计: TypeScript SDK with multi-agent and realtime voice support.

适合: Web, Node and voice applications.

注意: Smaller ecosystem than the Python package and fast API churn.

[GitHub](#)

R03 | openai/codex | coding agent

快照: Rust; Apache-2.0; 106,864 stars; created 2025-04; active

设计: Terminal coding harness with sandboxed command execution, repository context and iterative edits.

适合: Local and remote software engineering workflows.

注意: Model-specific behavior and large operational surface; measure full harness, not model alone.

[GitHub](#)

R04 | anthropics/claude-code | coding agent

快照: Python; proprietary/see repo; 142,023 stars; created 2025-02; active

设计: Terminal agent with file, shell, git, subagent, hook, skill and compaction primitives.

适合: Long-running coding and general computer work.

注意: Client repository visibility does not expose the full hosted model or control plane.

[GitHub](#)

R05 | anthropics/claude-agent-sdk-python | SDK/harness

快照: Python; MIT; 7,935 stars; created 2025-06; active

设计: Generalizes Claude Code's harness primitives to application agents.

适合: Agents that need a computer, subagents, hooks and permission controls.

注意: Closely tied to Claude behaviors and platform evolution.

[GitHub](#)

R06 | google/adk-python | SDK/runtime

快照: Python; Apache-2.0; 21,194 stars; created 2025-04; active

设计: Code-first agents, tools, sessions, memory, artifacts, evaluation and deployment.

适合: Google Cloud/Gemini teams needing multi-agent and managed deployment.

注意: Broad abstraction surface can obscure runtime cost and state ownership.

[GitHub](#)

R07 | google/adk-js | SDK/runtime

快照: TypeScript; Apache-2.0; 1,354 stars; created 2025-08; active

设计: TypeScript implementation of ADK's agent and deployment abstractions.

适合: Node teams aligning with ADK architecture.

注意: Newer and less mature than the Python implementation.

[GitHub](#)

R08 | a2aproject/A2A | protocol

快照: Shell/spec; Apache-2.0; 25,423 stars; created 2025-03; active

设计: Protocol and SDK ecosystem for agent discovery, messaging, tasks and artifacts across vendors.

适合: Cross-organization agent interoperability.

注意: Protocol compatibility does not guarantee semantic capability or trust.

[GitHub](#)

R09 | microsoft/autogen | multi-agent framework

快照: Python; CC-BY-4.0 metadata; 60,525 stars; created 2023-08; active

设计: AgentChat abstractions over an event-driven core and extensions.

适合: Research and conversational multi-agent orchestration.

注意: Large version transitions and licensing/component details need review.

[GitHub](#)

R10 | microsoft/agent-lightning | training

快照: Python; MIT; 17,520 stars; created 2025-06; active

设计: Decouples agent execution and RL training through observability and trajectory transitions.

适合: Fine-tuning existing agents without rewriting their orchestration.

注意: Reward design, rollout cost and credit assignment remain difficult.

[GitHub](#)

R11 | microsoft/magentic-ui | computer-use

快照: Python; MIT; 10,069 stars; created 2025-05; active

设计: Human-centered browser and file agent built on AutoGen/Magentic patterns.

适合: Research on oversight and interactive web automation.

注意: Live web fragility and browser security remain central.

[GitHub](#)

R12 | microsoft/Orchard | training/runtime

快照: mixed; MIT; 478 stars; created 2026-04; active

设计: Kubernetes environment service and recipes for SWE, GUI and open agent harness training.

适合: Large-scale rollout, distillation, RL and evaluation infrastructure.

注意: Young project with meaningful operational complexity.

[GitHub](#)

R13 | langchain-ai/langgraph | durable runtime

快照: Python; MIT; 40,040 stars; created 2023-08; active

设计: Graph/state runtime with durable execution, persistence, streaming and human-in-the-loop.

适合: Stateful production workflows requiring recovery and explicit control flow.

注意: Graph complexity can grow into application-specific orchestration debt.

[GitHub](#)

R14 | langchain-ai/deepagents | agent harness

快照: Python; MIT; 27,949 stars; created 2025-07; active

设计: Batteries-included harness adding planning, file state, subagents and memory on LangGraph.

适合: Long-horizon research and coding prototypes.

注意: Convenience layers inherit dependencies and defaults from the LangChain stack.

[GitHub](#)

R15 | crewAIInc/crewAI | multi-agent framework

快照: Python; MIT; 57,338 stars; created 2023-10; active

设计: Role/task/crew and flow abstractions for collaborative automation.

适合: Business workflows where roles and sequence are easy to express.

注意: Role-playing can become cosmetic unless information and responsibility boundaries are real.

[GitHub](#)

R16 | pydantic/pydantic-ai | typed SDK

快照: Python; MIT; 19,399 stars; created 2024-06; active

设计: Typed model and tool interfaces integrated with validation, evals and observability.

适合: Python teams prioritizing type safety and model portability.

注意: A typed boundary catches schema errors, not semantic or planning failures.

[GitHub](#)

R17 | huggingface/smolagents | minimal SDK

快照: Python; Apache-2.0; 28,890 stars; created 2024-12; active

设计: Minimal agents that often express plans and tool composition as code.

适合: Readable experiments and code-agent research.

注意: Small core does not provide a production control plane or isolation by itself.

[GitHub](#)

R18 | browser-use/browser-use | web agent

快照: Python; MIT; 109,782 stars; created 2024-10; active

设计: Browser abstraction plus agent loop over Playwright-like interactions.

适合: Web automation and browser-agent prototyping.

注意: Site change, anti-bot behavior, credentials and prompt injection create operational risk.

[GitHub](#)

R19 | OpenHands/OpenHands | coding platform

快照: TypeScript; MIT; 84,512 stars; created 2024-03; active

设计: Full coding-agent platform with isolated runtime, sessions, UI and extensible agents.

适合: Multi-user hosted software-development agents.

注意: Deployment complexity is much larger than a local CLI harness.

[GitHub](#)

R20 | SWE-agent/SWE-agent | coding/eval

快照: Python; MIT; 20,079 stars; created 2024-04; active

设计: Research harness for GitHub issue resolution and other terminal tasks.

适合: Transparent coding-agent research and benchmark baselines.

注意: Benchmark performance is sensitive to environment and task contamination.

[GitHub](#)

R21 | mem0ai/mem0 | memory layer

快照: Python; Apache-2.0; 63,619 stars; created 2023-06; active

设计: Cross-application memory extraction, storage and retrieval service.

适合: Personalization and persistent user facts across sessions.

注意: Memory quality, deletion semantics and provenance must be evaluated per use case.

[GitHub](#)

R22 | getzep/graphiti | temporal graph memory

快照: Python; Apache-2.0; 30,104 stars; created 2024-08; active

设计: Real-time temporal knowledge graph construction and retrieval for agents.

适合: Entity-centric histories with changing facts.

注意: Graph extraction errors can appear authoritative and propagate widely.

[GitHub](#)

R23 | letta-ai/letta | stateful agent platform

快照: mixed; Apache-2.0; 24,307 stars; created 2023-10; active

设计: Long-lived agents with editable memory, tools and state management.

适合: Research and products centered on persistent agent identity/state.

注意: Long-term state creates privacy, drift and lifecycle obligations.

[GitHub](#)

R24 | agno-agi/agno | agent platform

快照: Python; Apache-2.0; 41,790 stars; created 2022-05; active

设计: Broad toolkit for building, running and managing agents and teams.

适合: Teams wanting many integrations under one Python platform.

注意: Breadth increases dependency and abstraction risk; choose a minimal subset.

[GitHub](#)

R25 | camel-ai/camel | multi-agent research

快照: Python; Apache-2.0; 17,610 stars; created 2023-03; active

设计: Communicative-agent and society abstractions plus research environments.

适合: Multi-agent research, simulations and data generation.

注意: Simulation success does not imply robust operational collaboration.

[GitHub](#)

R26 | QwenLM/Qwen-Agent | SDK/app

快照: Python; Apache-2.0; 16,992 stars; created 2023-09; active

设计: Function calling, MCP, code interpreter, RAG and browser extensions for Qwen models.

适合: Qwen-centric local or enterprise agent applications.

注意: Recent push cadence was lower at the snapshot; verify current compatibility.

[GitHub](#)

R27 | Alibaba-NLP/DeepResearch | research agent

快照: Python; Apache-2.0; 19,851 stars; created 2025-01; active

设计: Open deep-research agent and training/evaluation assets for information seeking.

适合: Reproducible web research agents and open models.

注意: Live search, citation quality and site access dominate practical reliability.

[GitHub](#)

R28 | modelcontextprotocol/servers | tool ecosystem

快照: TypeScript; see repo; 89,696 stars; created 2024-11; active

设计: Reference and community MCP servers exposing external capabilities.

适合: Learning tool contracts and bootstrapping integrations.

注意: Community tools are supply-chain inputs and require provenance and permission review.

[GitHub](#)

R29 | modelcontextprotocol/modelcontextprotocol | protocol/spec

快照: TypeScript; see repo; 8,998 stars; created 2024-09; active

设计: Specification and documentation for client-server context and tool exchange.

适合: Protocol implementers and platform integration teams.

注意: MCP standardizes transport/shape, not trust, semantics or policy.

[GitHub](#)

R30 | FoundationAgents/MetaGPT | multi-agent framework

快照: Python; MIT; 69,901 stars; created 2023-06; active

设计: Software-company role workflow and natural-language programming patterns.

适合: Educational multi-agent software-process experiments.

注意: Role simulation and benchmark claims need controlled comparison with simpler agents.

[GitHub](#)

R31 | Aider-AI/aider | coding agent

快照: Python; Apache-2.0; 48,333 stars; created 2023-05; active

设计: Git-aware terminal pair programmer with repository maps and edit formats.

适合: Interactive coding with explicit human steering.

注意: Less suited to unattended multi-user control-plane workloads.

[GitHub](#)

R32 | cline/cline | coding agent

快照: TypeScript; Apache-2.0; 66,499 stars; created 2024-07; active

设计: IDE, CLI and SDK agent with file, terminal, browser and approval flows.

适合: Developer-in-the-loop work inside editors.

注意: Extension permissions and provider configuration expand the attack surface.

[GitHub](#)

R33 | continuedev/continue | coding agent

快照: TypeScript; Apache-2.0; 35,546 stars; created 2023-05; active

设计: Open coding agent and editor integrations with model/provider flexibility.

适合: Teams needing self-hosted or model-flexible coding assistance.

注意: Configuration flexibility transfers integration and evaluation work to the operator.

[GitHub](#)

R34 | google-gemini/gemini-cli | coding/general agent

快照: TypeScript; Apache-2.0; 106,586 stars; created 2025-04; active

设计: Terminal agent with Gemini models, tools, MCP and extensions.

适合: Google-centric development and general terminal workflows.

注意: Provider coupling and broad terminal permissions need explicit containment.

[GitHub](#)

R35 | BloopAI/vibe-kanban | orchestrator/UI

快照: Rust; Apache-2.0; 27,856 stars; created 2025-06; active

设计: Task-board orchestration over multiple external coding agents.

适合: Human supervision of parallel coding work.

注意: It coordinates harnesses but does not eliminate merge, verification or context problems.

[GitHub](#)

R36 | humanlayer/humanlayer | coding workflow
快照: TypeScript; see repo; 11,296 stars; created 2024-08; active
设计: Human-in-the-loop workflow and context engineering for complex codebases.
适合: High-context changes needing planning and review checkpoints.
注意: Process quality depends on disciplined task decomposition and review capacity.
[GitHub](#)

7. 局限、复核和更新建议

研究局限

- 本系列是结构化深度调查，不是穷尽过去一年所有Agent论文。筛选强调对系统设计和实证判断的贡献。
- 许多论文只依据摘要和公开页面做高层编码；对关键主张保留原文链接，重要决策应阅读全文和代码。
- 厂商博客具有第一手工程价值，也存在选择性披露。报告把它们与独立论文/基准分开解释。
- 仓库stars、issues和状态会变化；许可证字段也可能是仓库元数据而非所有子组件许可。采用前必须做当前版本审计。
- 模型、harness和基准变化速度很快，任何具体排行榜数字都应带日期。

复核清单

- 核对论文版本(v1/v2/accepted)以及是否有后续修订。
- 检查代码、数据、容器和评测脚本能否运行。
- 重新计算关键分数并记录模型、harness、环境和成本。
- 阅读失败样例，而不是只读均值和SOTA表。
- 对安全、医疗、法律和金融结论咨询相应专业人员。
- 在组织自己的任务集上复现，不直接迁移论文结论。

建议更新节奏

论文和仓库索引建议每月增量更新，核心结论每季度复核，安全威胁与协议版本按发布即时更新。更新时保留旧快照，记录新增、撤回、复现失败、仓库迁移和指标定义变化。只有这样，年度研究才能成为知识资产，而不是一次性报告。

索引用途
本卷提供可追溯入口。任何重要结论都应回到链接原文，结合具体模型、harness、环境、工具和预算再判断。系列其他各卷是分析层，本卷是证据账本。